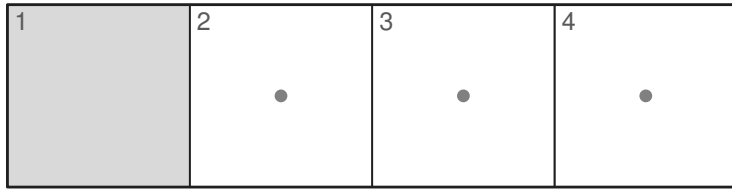
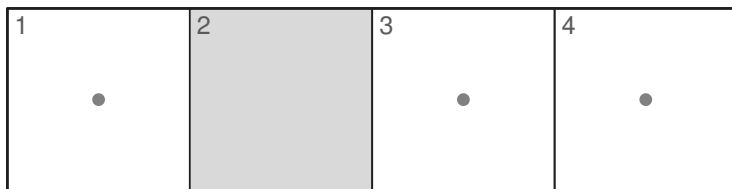


Problem 1: Put a penny on every square except the grey one. A penny jumps over the penny next to it into the empty square right behind it, and the penny it jumped over comes off the strip. Jump until no jump is left. For each strip, write how many pennies are left and, if only one is left, the number of its square. Do all four strips.



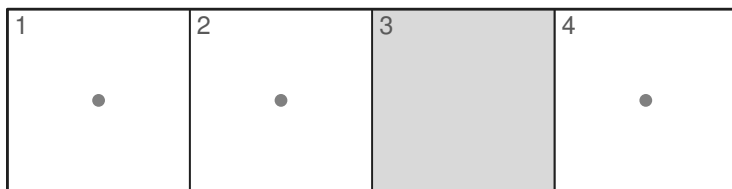
pennies left

last square



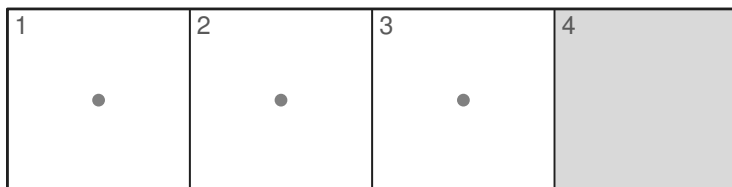
pennies left

last square



pennies left

last square

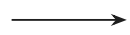
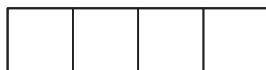
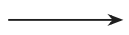


pennies left

last square

Problem 2: Write a jump as three numbers: $3>2>1$ means the penny on square 3 jumps over square 2 and lands on square 1. For the strip with the hole on square 1, fill in the small strips with every game that can be played from the start (a dot for each penny), and write each jump on the line under its arrow. Do the same for the hole on square 2. You may not need every strip. Write how many different games each strip has.

hole on 1



games: _____

hole on 2



games: _____

Problem 3: These strips have five squares. Play every strip as in Problem 1 and fill in the boxes. Play a strip again if you think you can leave fewer pennies. Could any of the five be finished with one penny?

1	2	3	4	5
	•	•	•	•

pennies left last square

1	2	3	4	5
•		•	•	•

pennies left last square

1	2	3	4	5
•	•		•	•

pennies left last square

1	2	3	4	5
•	•	•		•

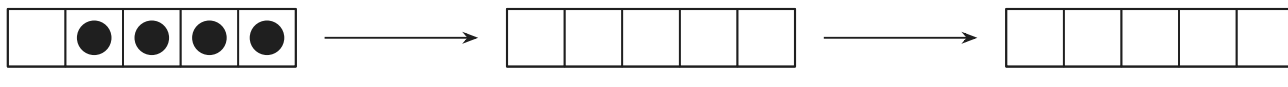
pennies left last square

1	2	3	4	5
•	•	•	•	

pennies left last square

Problem 4: Fill in every game on the strip of five for the hole on square 1, then for the hole on square 2, then for the hole on square 3. Write each jump on the line under or beside its arrow. Holes 4 and 5 are holes 2 and 1 seen from the other end. How many games are there in all? How many pennies are left at the end of each game?

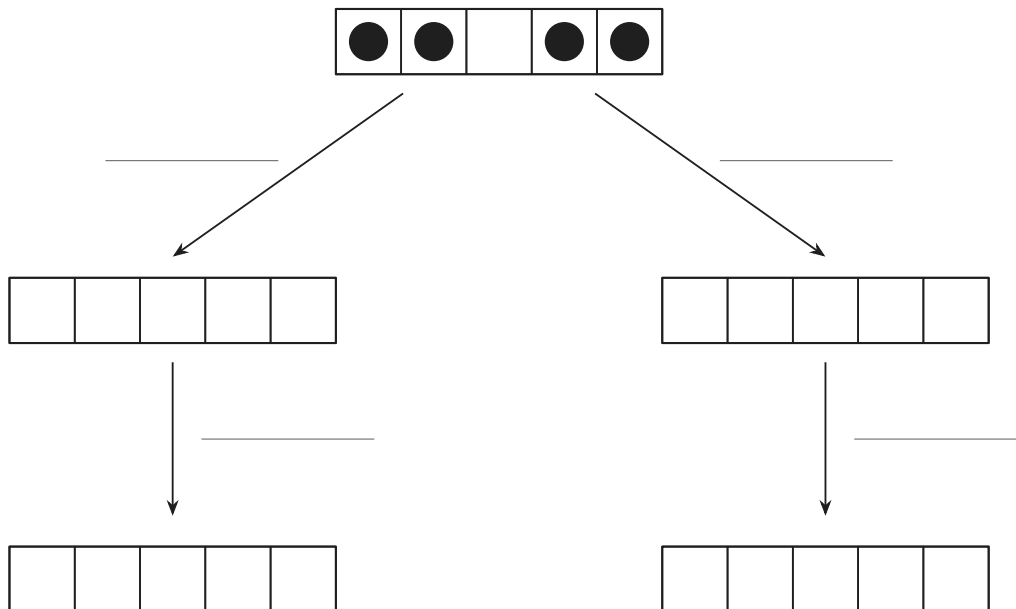
hole on 1



hole on 2



hole on 3



games in all

pennies left at the end of each game

Problem 5: Leave square 1 empty and put pennies on the other five squares. Jump until no jump is left, and fill in the row for hole 1 in the table. Then do hole 2, and so on up to hole 6. Play a hole again if you think you can leave fewer pennies.

1	2	3	4	5	6
•	•	•	•	•	•

hole on square	pennies left	last penny on square
1		
2		
3		
4		
5		
6		

Problem 6: Do the same on the strip of seven squares, for holes 1 to 4. Holes 5, 6 and 7 are holes 3, 2 and 1 seen from the other end.

1	2	3	4	5	6	7
•	•	•	•	•	•	•

hole on square	pennies left	last penny on square
1		
2		
3		
4		

Problem 7: Do the same on the strip of eight squares, for holes 1 to 4. Holes 5 to 8 are holes 4 to 1 seen from the other end. For every hole that can be finished, play it again and see whether the last penny can end on a different square; write down every ending square you find. Then look back at Problems 1 to 7 and fill in the three lines.

1	•
2	•
3	•
4	•
5	•
6	•
7	•
8	•

hole on square	pennies left	last penny on square
1		
2		
3		
4		

strip lengths that could be finished with one penny:

the holes that worked on them:

strip lengths that never could be finished:

Problem 8: Colour each square of this strip: R means red, B means blue, G means green. Put pennies on every square except 2. Count the pennies on red squares, on blue squares and on green squares, and write the three numbers in the row marked start. Make one jump and count again. Keep going until the game ends, one row for each jump. In the last column write the total number of pennies on the strip.

1	2	3	4	5	6
•		•	•	•	•
R	B	G	R	B	G

	red	blue	green	total
start				
after jump 1				
after jump 2				
after jump 3				
after jump 4				

Problem 9: Colour this strip the same way. Put pennies on every square except 5 and play the game again, counting after every jump.

1	2	3	4	5	6
•	•	•	•		•
R	B	G	R	B	G

	red	blue	green	total
start				
after jump 1				
after jump 2				
after jump 3				
after jump 4				

Problem 10: Colour these strips the same way. Under each strip write how many pennies would stand on red, on blue and on green at the start, when every square except the grey one has a penny. Then look at your boxes and tables from Problems 1 to 7 and mark each strip yes if it could be finished with one penny and no if it could not. For each yes strip, also write the colour of the square where the last penny ended.

1	2	3	4
R	B	G	R

red ____ blue ____ green ____

finished: yes / no colour of the last square: _____

1	2	3	4
R	B	G	R

red ____ blue ____ green ____

finished: yes / no colour of the last square: _____

1	2	3	4	5
R	B	G	R	B

red ____ blue ____ green ____

finished: yes / no colour of the last square: _____

1	2	3	4	5
R	B	G	R	B

red ____ blue ____ green ____

finished: yes / no colour of the last square: _____

1	2	3	4	5	6	7
R	B	G	R	B	G	R

red ____ blue ____ green ____

finished: yes / no colour of the last square: _____

1	2	3	4	5	6	7	8
R	B	G	R	B	G	R	B

red ____ blue ____ green ____

finished: yes / no colour of the last square: _____

1	2	3	4	5	6	7	8
R	B	G	R	B	G	R	B

red ____ blue ____ green ____

finished: yes / no colour of the last square: _____

1	2	3	4	5	6	7	8
R	B	G	R	B	G	R	B

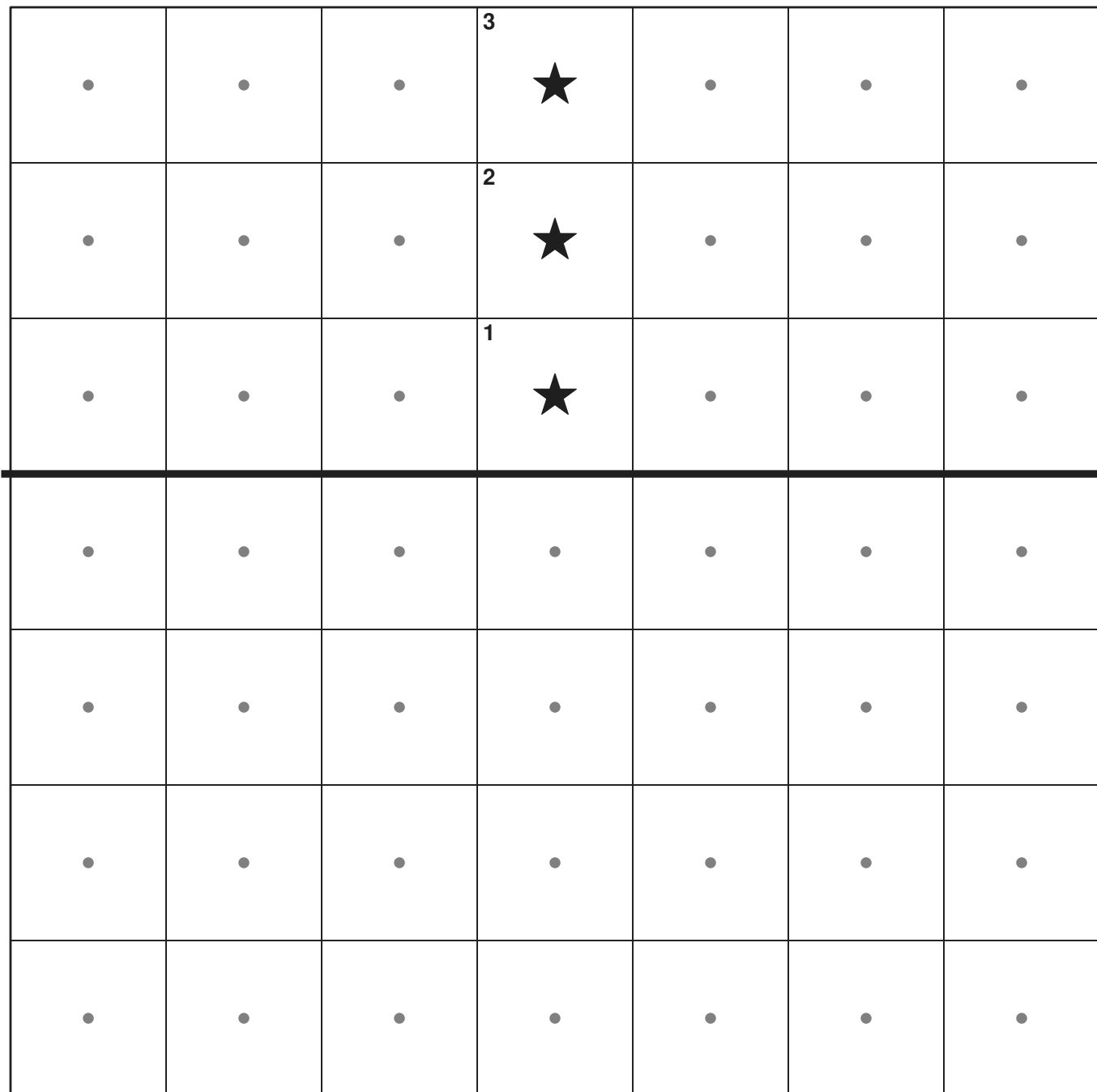
red ____ blue ____ green ____

finished: yes / no colour of the last square: _____

Problem 11: Use the colours to predict, before playing, what will happen on these strips. Write your prediction: cannot be finished, or can be finished and the last penny will be on a red, blue or green square. Then play each one on your strips from Problems 6 and 7 and write what happened.

strip	red, blue, green at the start	prediction	what happened
eight squares, hole on 6			
eight squares, hole on 5			
seven squares, hole on 7			
eight squares, hole on 8			

Problem 12: The pennies are soldiers. Soldiers start on squares below the thick line, never above it. They jump as before, along a row or a column, never slantwise. Get a soldier onto star 1 using as few soldiers as you can. Then get a soldier onto star 2 with as few soldiers as you can, and then star 3. Draw each smallest army on the boards of Problem 13.



Problem 13: Draw a dot for each soldier of your smallest army for star 1, for star 2 and for star 3, and write the number of soldiers under each board. Then write your guess for the smallest army that could reach a star four rows above the line.

star 1

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.	.	.	★	.	.	.
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star 2

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.	.	.	★	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.

star 3

.	.	.	★	.	.	.
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my guess for star 4

Problem 14: On this board draw your smallest army for star 2 again, a dot for each soldier. Then draw an arrow for each jump, from the square the soldier leaves to the square it lands on, and number the arrows 1, 2, 3 in the order of the jumps.

.	.	.	.	.	.	.
.	.	.	★	.	.	.
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.	.	.	.	.	.	.
.	.	.	.	.	.	.

Problem 15: Put pennies on any of squares 1 to 6, as many as you like, and none past the thick line. Jump until no jump is left. In the table, draw a dot on each square that had a penny at the start, and write the number of the square farthest to the right that any penny reached. Do at least five different armies. What is the smallest army that gets a penny past the line? Can a penny ever reach square 8?

1	2	3	4	5	6		7	8	9
•	•	•	•	•	•		•	•	•

army at the start

--	--	--	--	--	--	--	--	--	--

farthest square

--

--	--	--	--	--	--	--	--	--	--

--

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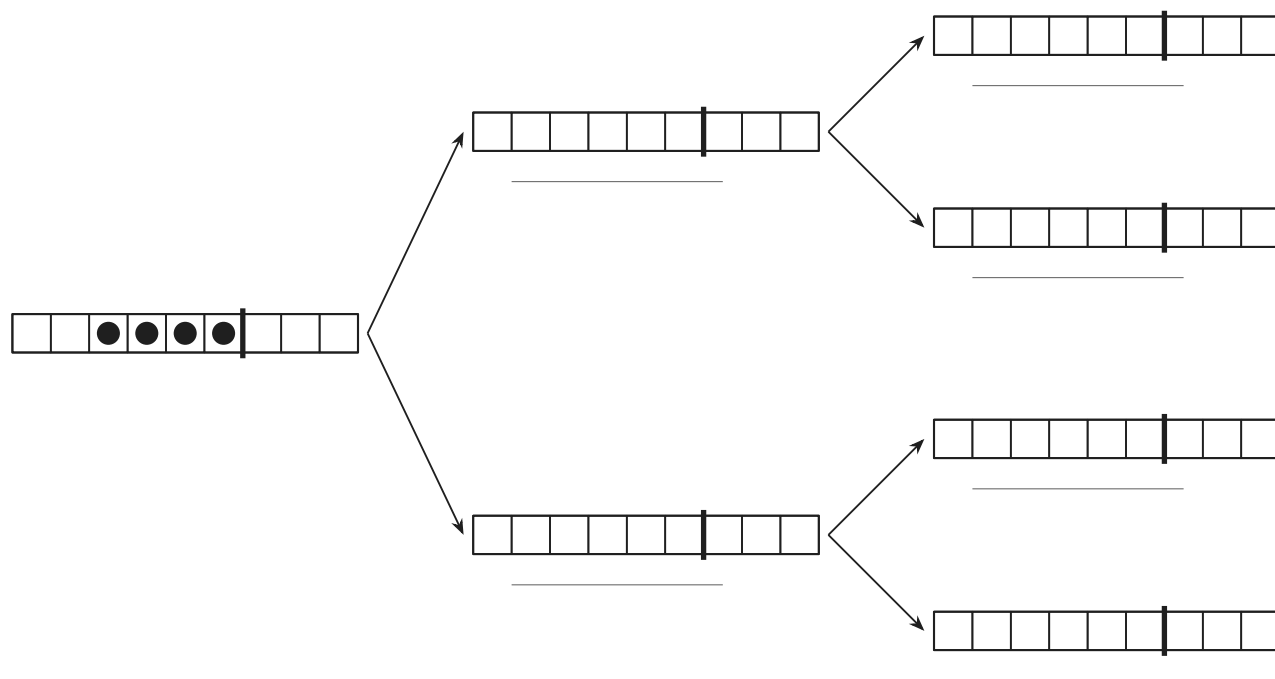
smallest army that gets past the line

--

can a penny reach square 8?

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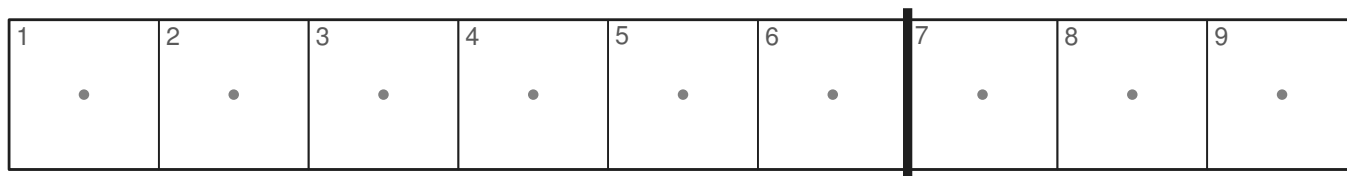
Problem 16: Put pennies on squares 3, 4, 5 and 6 only. Fill in every game that can be played on the small strips, and write each jump on its line. How many games are there? What is the farthest square a penny reaches in any of them?



games in all

farthest square in any game

Problem 17: Put pennies on all six squares 1 to 6. Play the game. How many different first jumps are there? Is there any choice at all before the game ends?



different first jumps

jumps where there was a choice

Problem 18: Each square of these strips has a number. Put pennies on the four squares to the left of the thick line. Add up the numbers under the pennies and write the total in the first box. Play the game; after every jump, add up the numbers under the pennies again and write the new total. Leave a column empty if the game ends before it. Can a penny reach the star? Compare the star's number with your totals.

1	1	2	3	5	8 ★
---	---	---	---	---	-----

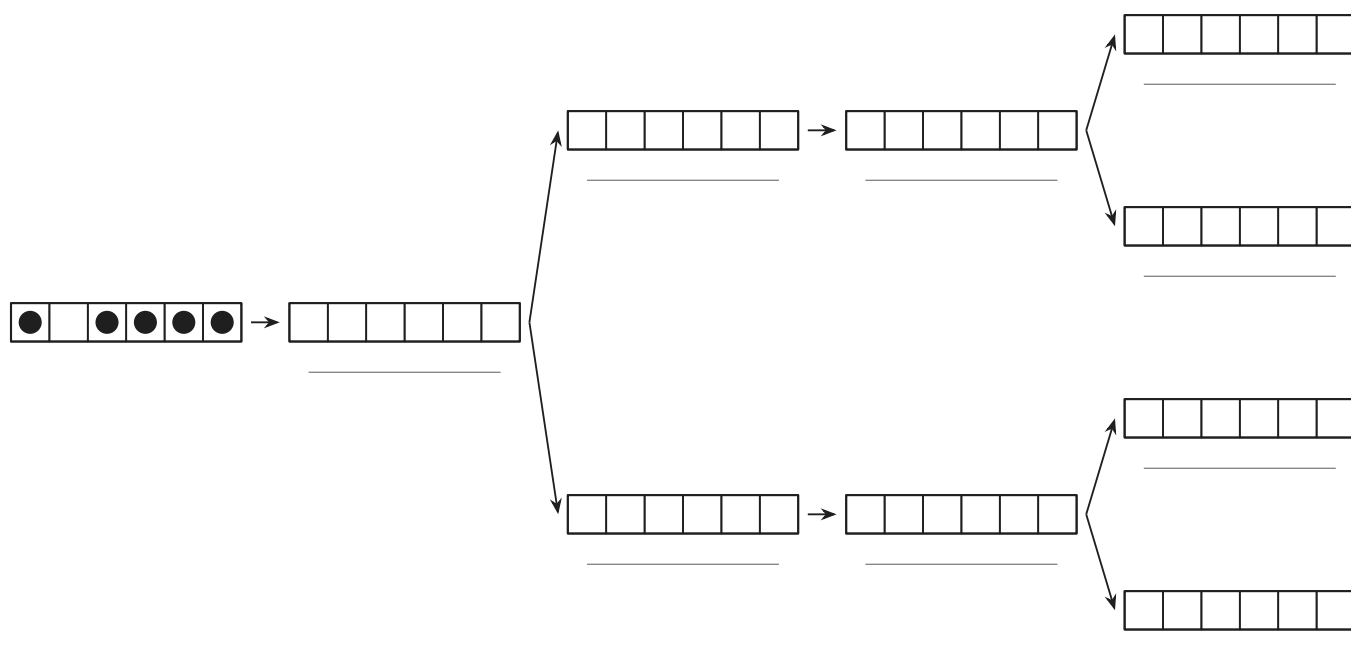
start	after jump 1	after jump 2	after jump 3	star

Problem 19: Now five pennies, on the five squares to the left of the thick line. Add up before you start and after every jump. Can a penny reach the star?

1	1	2	3	5	8	13 ★
---	---	---	---	---	---	------

start	after jump 1	after jump 2	after jump 3	after jump 4	star

Problem 20: The strip of six with the hole on square 2 can be finished in more than one way. Fill in the small strips with every possible game, writing each jump on its line. How many different winning games are there?

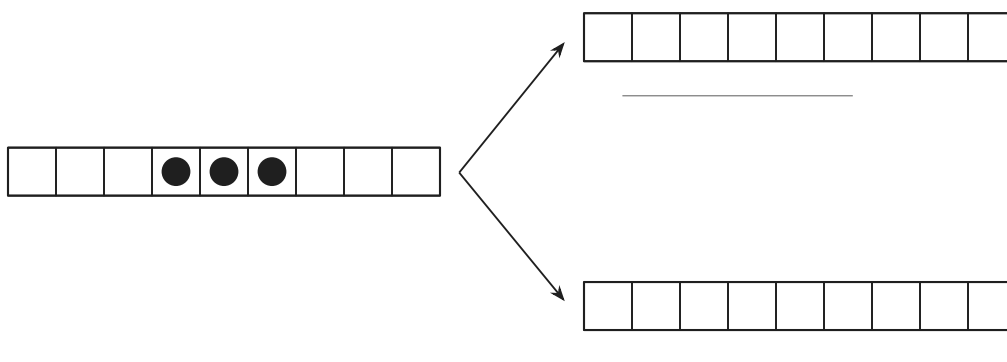


winning games

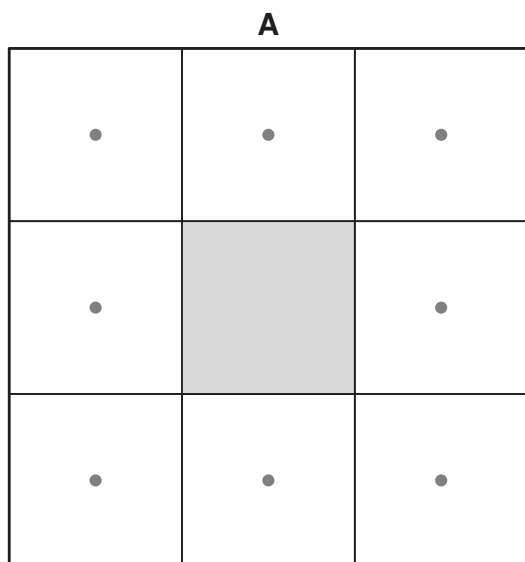
Problem 21: Put pennies on squares 4, 5 and 6 only, a block of three. Jump until no jump is left, in any direction; jumps may leave the block. Write the fewest pennies you can leave. Then do a block of four (squares 4 to 7), a block of five (3 to 7) and a block of six (3 to 8). Then fill in every game for the block of three on the small strips.

1	2	3	4	5	6	7	8	9
---	---	---	---	---	---	---	---	---

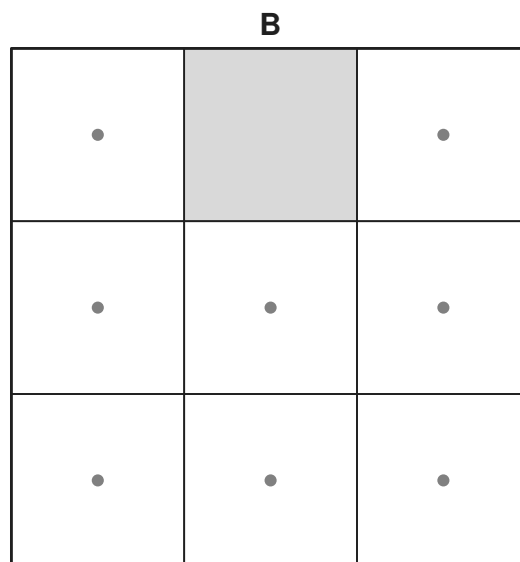
block of	3	4	5	6
fewest pennies left				



Problem 22: Three square boards. A jump goes left, right, up or down, never slantwise. Put pennies on every square except the grey one. On board A, count how many different jumps are possible from the start. On board B, count how many are possible from the start, then jump until no jump is left. On board C, jump until no jump is left and write how many pennies remain. Play board C three times and write the fewest you managed.

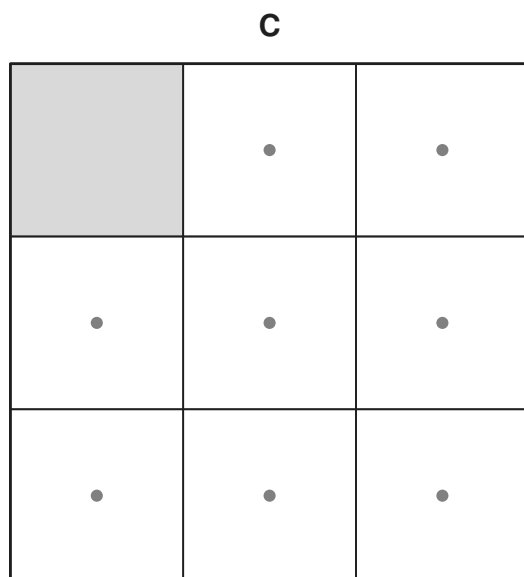


jumps at the start



jumps at the start

pennies left



pennies left, game 1

pennies left, game 2

pennies left, game 3

fewest

Problem 23: Leave square A empty and put pennies on the other eleven squares. Jump until no jump is left. In the table write how many pennies are left and, if only one is left, the letter of its square. Then do the same with the hole on B, on E and on F. For a hole that can be finished, play it again and try to make the last penny end on a different square; write down every ending square you find.

A •	B •	C •	D •
E •	F •	G •	H •
I •	J •	K •	L •

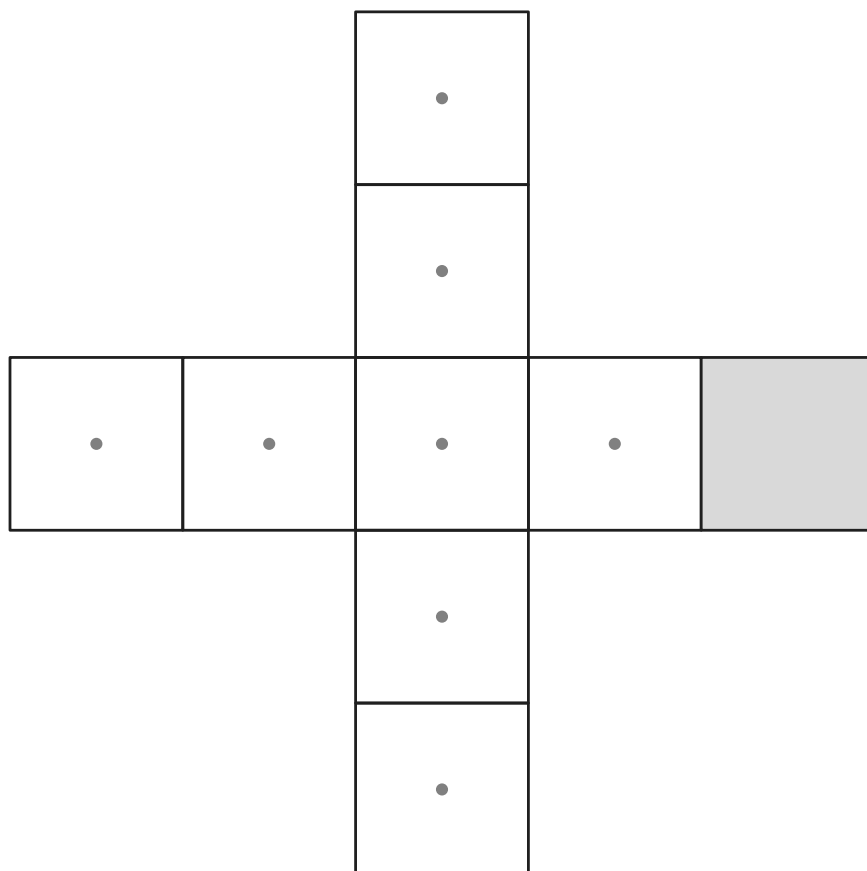
hole on square	pennies left	last penny on square
A		
B		
E		
F		

Problem 24: This board has fifteen squares. Leave square C empty and put pennies on the other fourteen. Jump until no jump is left. Write how many pennies are left and, if one, its square. Then try the hole on A, on B and on H. Which of the four holes can be finished?

A	B	C	D	E
F	G	H	I	J
K	L	M	N	O

hole on square	pennies left	last penny on square
C		
A		
B		
H		

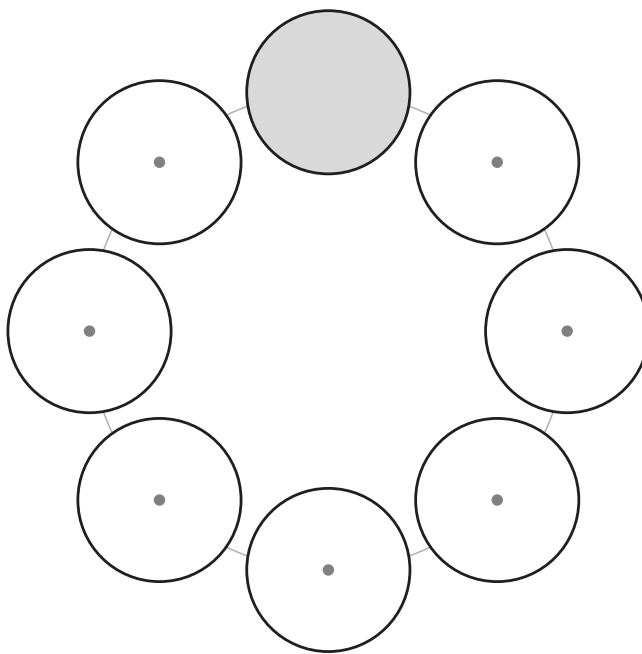
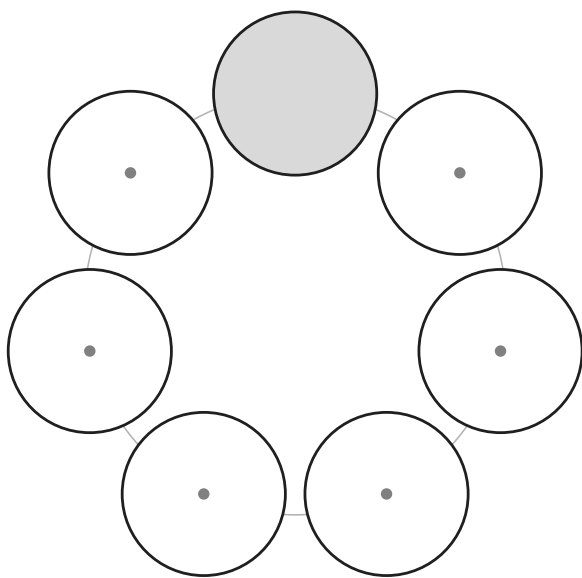
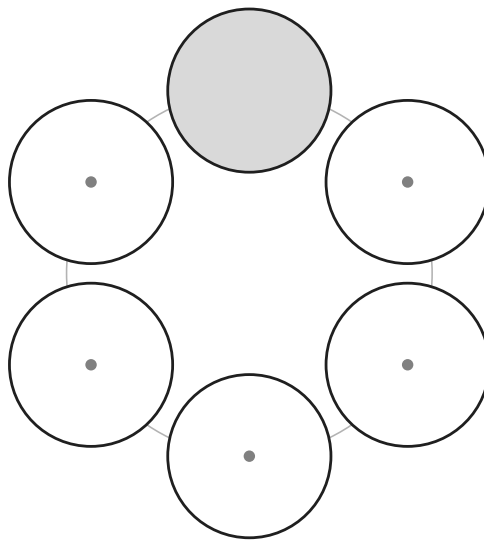
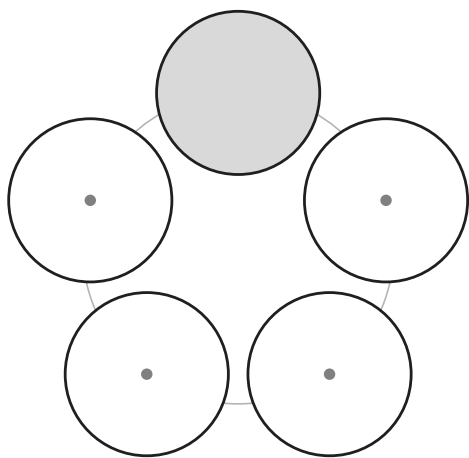
Problem 25: Put pennies on every square of the cross except the grey one. Jump until no jump is left. Play several times and try to leave as few pennies as you can. Then move the hole to the centre square and try again.



fewest left, hole at the end of an arm

fewest left, hole in the centre

Problem 26: Cells around a ring. Put a penny on every cell except the grey one. A penny jumps over the penny next to it into the empty cell just beyond, going around the ring. On each ring, jump until no jump is left and write how many pennies are left; play it again and try to leave fewer. Which rings can be finished with one penny?



ring of	5	6	7	8
fewest pennies left				